An End-to-End Pipeline for Glyph Detection & Writer Identification on Ancient Greek Papyri

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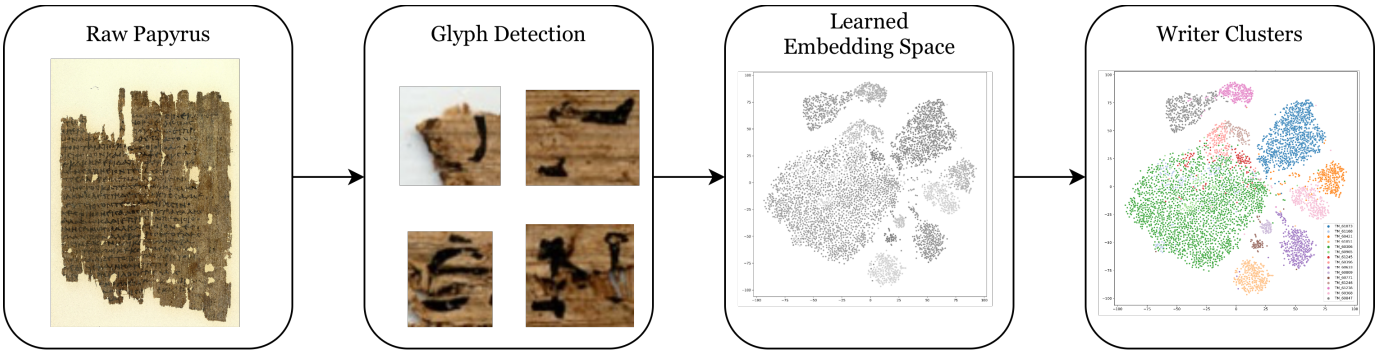


Fig. 1. Project Overview: The complete pipeline from raw papyrus input to writer identification.

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***Abstract*—Identifying writer from ancient Greek papyri is not mainly about understanding the text, it is about analyzing handwriting styles unique to each person. This is a challenging computer vision task, since documents are often damaged, torn, or incomplete. Besides that, there is no extensive labeled dataset available. Here, we describe building a complete system that turns unprocessed image scans into identified writers.**

To find small glyphs in large, high-resolution images, we built a robust YOLOv8 tool that uses Slicing Aided Hyper Inference (SAHI). When it came to recognizing characters, we first tried SimCLR, which works without labels. We later shifted instead to a Supervised Contrastive setup, for a more robust evaluation method. We used ResNet50 paired with Triplet Margin Loss to clearly distinguish writer embeddings. This refined system reaches a mean Average Precision (mAP) of 93.73%, tested on previously unseen writers.

***Index Terms*—Papyri, Writer Identification, Computational Paleography, SimCLR, Simple-VLAC**

I. Introduction

The study of ancient handwriting, also known as paleography, serves the main method used to determine age and verify origin in history texts without clear data labels. Within this particular field in Greek papyrology, pinpointing a single scribe relies on distinct handwriting traits/styles, and not just sorting into categories. This helps researchers piece together scattered ancient archives, follow how knowledge spread across regions from key works such as the Homeric poems, extracting their meaning through close reading and contextual analysis. Yet standard paleographic analysis relies heavily on the visual memory alongside personal insight from researchers, which makes it hard to copy widely.

Automated Writer Identification (WI) attempts to provide quantitative measures. Although WI has been widely adopted, using this method on ancient papyri documents instead presents several distinct difficulties. These material irregularities frequently interfere with standard computer vision pipelines, which struggle to distinguish ink from the textured background.

A. The Challenges of Papyrology

The main challenge in this domain is the condition of the source material. Since papyri are bio-materials that have survived for years, they are often riddled with lacunae (holes), worm tracks, and fiber peeling. Because of this, the background texture often visually overpowers the faint ink strokes, creating a low signal-to-noise ratio. For this reason, any effective solution cannot rely on macro-structural features like line spacing or page margins, and must operate at the granular level of individual glyphs.

Furthermore, the digital scans of these documents brings in added distortions, including inconsistent illumination, sensor artifacts, and cropping biases. Deep learning models are prone to ”shortcut learning”, wherein they lazily exploiting these digital artifacts to identify the image file rather than the

writer's hand. This is a phenomenon we diagnose and mitigate moving forward in this project.

B. Research Path and Contributions

The objective of the project was to build a stable, end-to-end pipeline for character-based writer identification on the *HomerComp* dataset. This required a two-stage approach handling distinct computer vision challenges:

1) Glyph Detection (Task A): Detecting individual Greek glyphs within high-resolution raw images without losing fine details from downscaling.
2) Writer Identification (Task B): Developing a metric learning objective that extracts writer-specific embeddings while remaining unaffected by papyrus damage, through contrastive learning.

This report documents the step-by-step progression of the project, placing specific emphasis on the engineering hurdles encountered regarding data integrity and the solutions developed to overcome them. We explain how early benchmarks resulted in below-average results because of flawed data patterns, and the identification of "source bias" within initial self-supervised models.

Our main work includes:

1) Robust Detection Pipeline: We adapted YOLOv8 and SAHI to handle high resolution manuscripts without drawing out key details.
2) Methodological Pivot: We document the transition from SimCLR (which learned artifacts) to Triplet Loss (which imposed stylistic training), presenting a real-world example in choosing a model when working with noisy data.
3) State-of-the-Art Performance: By implementing a character-specific aggregation strategy (Simple-VLAC), we achieved 93.73% mAP result. This is clearly higher than global averaging methods.

II. Related Work

Our approach builds upon the ICDAR contest, while adapting state-of-the-art methods initially developed for facial recognition methods to the domain of paleographic studies.

A. Evolution of the Challenge

The background of this project is shaped by the ICDAR 2019 Competition on Image Retrieval for Historical Handwritten Documents [2]. The ICDAR 2019 event highlighted the severe challenges that come with pairing writers within scattered ancient handwritten texts. This contest showed how while texture-based methods (like SIFT, LBP, or HOG) are strong, they still have some trouble distinguishing background noise from ink strokes on degraded documents.

Using this idea, Seuret et al. (2023) proposed the HomerComp dataset and the ICDAR 2023 Competition on Detection and Recognition of Greek Letters on Papyri [3]. This formalized the basis for the clear two-step pipeline we used: initially, detect the glyphs before identifying the writer. This division of tasks is crucial for dealing with fragmented texts. Our initial approach used PapyTwin architecture, the winner of the 2023 competition, as a starting point. But this was clearly evolved later in the training approach to tackle generalization issues.

B. Approaches to Metric Learning

A core challenge in identifying writers is the shortage of massive labeled datasets required for standard classification. To tackle this issue, new studies have started using metric learning, in which the aim is to develop an embedded environment instead of a decision boundary.

We initially explored Self-Supervised Learning (SSL), particularly SimCLR [5], where representations are learned through generalizing across augmented versions of the same images. Though useful when labels are limited, our approach discovered SimCLR mostly adapts to shape changes through contrastive training- changes that don't necessarily ensure separability among various writers using comparable handwriting techniques (see Section IV).

As a result, our ideas came from a study by Schroff et. al. on FaceNet [6]. Their work demonstrated how image data can be transformed straight into Euclidean space using coordinate-based methods via Triplet Margin Loss. This works better than intermediate classification layers used in validation processes. By adapting this concept from face recognition to writer identification, our model pushes the network to group together glyphs belonging to the same writer (inter-class distance) and separating different glyphs among groups (intra-class distance), regardless of the particular character being written.

C. Character-Specific Aggregation

Lastly, our approach to combining data draws on recent developments in computational paleography. This was drawn from the "KaiRacters" paper proposed by Peer et al. [4]. They suggested that a writer's handwriting style is not uniformly distributed across the letters; rather, complex allographs (like ε or ξ) carry significantly more discriminative weight than simple strokes. Testing on the *HomerComp* dataset helped us validate this hypothesis, confirming that aggregating features from these *"strong"* characters yields higher retrieval accuracy than indiscriminate global averaging.

III. Dataset and Pre-processing

The basis of this project is the *HomerComp* dataset [3]. Rather than using processed images, this dataset provides high-resolution scans directly from papyri. This resolution is crucial, since fine ink spread and paper structure are visible only at such high resolutions.

A. Dataset Characteristics

The ground truth labels for identifying writers are provided alongside the data split into Trismegistos numbers, known as Trismegistos (TM) IDs. These IDs are uniquely assigned to distinguish one ancient document from another. Each identifier links exclusively to a single document in this setting. The TM ID serves as the class label, taking it that each document matches just one writer for these fragments.

The raw data included labels for 140 distinct items scribes. Although at first glance, we noticed an extended tail distribution: several scribes appeared just a few times made up of fragmented characters, usually occluded or distorted. Training a metric learning model on this kind of sparse data would lead to unstable gradients. For this reason, we used a quality check, by eliminating every writer who lacks at least 10 strong glyphs samples.

The final curated dataset consists of 34,478 glyphs distributed among 94 distinct writers. Class proportions still showed uneven distribution, with 119 separate characters types; common glyphs such as alpha (α) occur many times, and uncommon ones have less than ten instances.

IV. Methodology for Glyph Detection (Task A)

Before identifying the writer, it's necessary to begin by solving the issue of localizing the handwritten text (Glyph).

A. Data split

To validate our approach, we applied writer-disjoint split:

- Training Set (80%): 75 writers. Used to train the feature encoder.
- Test Set (20%): 19 writers. Used exclusively for evaluation.

Importantly, the system *never sees* the test writers during training. This ensures that the reported metrics align with the model's capacity to pick up a generalized representation of the handwriting style, instead of just learning the particular habits of the training set.

B. YOLOv8 Model Selection

We selected the YOLOv8 architecture for its state-of-the-art balance between inference latency and recall. While newer iterations are available, the primary motivation for this decision was ecosystem compatibility, specifically regarding integration with SAHI. However, deploying a general-purpose object detector—typically optimized for distinct macro-objects like vehicles—to the domain of microscopic, degraded handwriting required iterative model selection.

1) Addressing Underfitting: We initially deployed the lightweight `YOLOv8-Nano` architecture (3.2M parameters), operating under the hypothesis that Greek glyphs are geometrically simple primitives that would not require a deep feature space. This assumption proved incorrect. We started testing using the `YOLOv8-Nano` model (3.2M parameters). The method didn't work as the training loss went down slowly, but then stopped stagnated quickly; meanwhile, mAP stayed close to 0%. Examining the failure showed that the the model had trouble telling apart morphologically similar characters, such as differentiating a δ from an α. The nano version just didn't have the ability to distinguish the fine-grained stroke details that were needed for this 119-category task. We scaled up the model by using `YOLOv8-Medium` model contains 25.9 million parameters. This higher capacity quickly fixed the underfitting issue, enabling the model to reach stability and achieve a validation mAP score of 12.7%.

2) Training Protocol: The final detector was trained using AdamW optimizer, with a starting learning rate of 0.01; due due to the higher VRAM needs of the Medium version, we limited the batch size to 8; training continued across 300 epochs using early stopping on the convergence of mAP50-95 metric.

C. Resolution Gap (SAHI)

A major technical challenge in digital papyrology is the large size of the source images, usually above 4000×4000 pixels. Typical deep learning workflows adjust input size to a set square shape (for example, 640×640). On this level, a tiny Greek a letter such as *iota* turns into a fuzzy cluster of three or four pixels, which destroys the information necessary for identifying writers.

To resolve this issue, Slicing Aided Hyper Inference (SAHI) was introduced [7]. Instead of scaling down the entire image, which can lose details, we analyze it using a moving 640×640 window that scans across the whole pag

- Overlap Method: We implemented a 20% overlap between adjacent tiles. This is essential to ensure that glyphs located at the boundaries of a slice remain intact, preventing false negatives caused by arbitrary cropping.
- Result Merging: Detections from these overlapping areas are mapped back to the global image coordinates and consolidated using *Non-Maximum Suppression* (NMS) to eliminate duplicates.

This pipeline ensures that each glyph is detected at it is actual resolution, preserving the fine-grained stroke details essential for Task B.

D. Correcting Class Mapping Inconsistencies

A key part of the development process involved spotting a hidden data error. When we first switched to identifying writers, model performance stayed poor - accuracy hovered around ($\approx 9\%$). Adjusting standard settings did nothing to fix it. Because of this, we paused work to manually inspect sample batches from training data.

Identifying the problem: We observed a systematic misalignment between images and their labels. For example, glyphs visually identifiable as "Theta" (θ) appeared to be repeatedly tagged as "Epsilon" (ϵ) instead.

Root Cause and Fix: We traced the error to the manifest creation code. This used a separate JSON file, provided in HomerComp, which assigns category IDs to labels. However the inner class numbering created by YOLO while learning didn't align with that of the source JSON format. The data pipeline was updated so it retrieves the trained YOLO and uses `model.names` internally to identify classes true value. The solution restored 34,478 valid, correctly labeled glyphs. That moment highlighted a lesson: *In deep learning, code correctness doesn't ensure data correctness.*

V. Methodology for Writer Identification (Task B)

After the glyphs are properly identified and tagged, the next task is to identify who it belongs to. One challenge involves

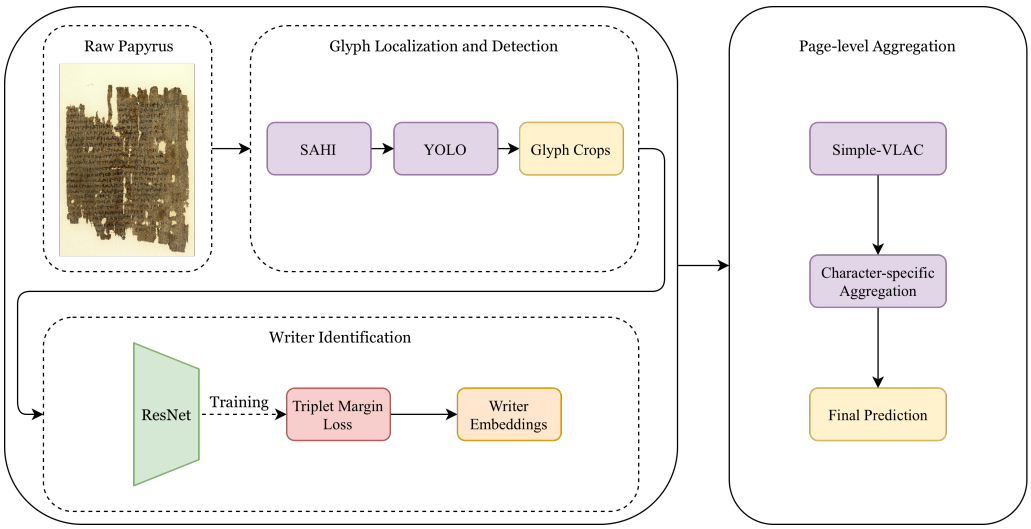


Fig. 2. Overview of our approach.

clustering glyphs belonging to same writer, and to separate clusters from different writers.

To set a clear starting point, we first expanded on the the structure behind PapyTwin, which won at ICDAR 2023 competition [3]. We used this ResNet-based setup but changed how we trained it. This was done by moving from binary classification pairs to contrastive learning, which helped in improving stability and generalization.

A. Exploration 1: Self-Supervised Learning (SimCLR)

We first tested *SimCLR* [5], a Self-Supervised Learning (SSL) approach that learns patterns without labels. this method used the *PapyTwin* model to capture meaningful features from large volumes of unannotated material using contrastive training.

1) Evaluation via Linear Probing: Since SimCLR is trained without labels, it does not inherently "know" which writer is which. To evaluate the quality of the learned features, we used the standard Linear Evaluation Protocol.

We froze the weights of the pre-trained ResNet encoder and trained a single linear classification layer (a "Linear Probe") on top of it using the writer labels and Cross-Entropy loss. If the linear probe achieves high accuracy, it proves that the encoder has successfully learned to cluster writers in the embedding space, thereby making the data "linearly separable."

2) Implementation Details: We trained a ResNet18 backbone to match two differentially augmented views of the same glyph instance. To prevent the model from overfitting, we implemented a strong augmentation methods:

- Random Resized Crop: Applied with a scale range of $[0.2, 1.0]$. This forces the model to detect a glyph based on local topological features (e.g., identifying an Alpha from only its crossbar) rather than global shape.
- Color Jitter: Random perturbations in brightness, contrast, and saturation ($strength = 0.4$) were used to simulate inconsistent lighting conditions and varying degrees of ink fading.
- Gaussian Blur: Applied with $p = 0.5$ to simulate the shallow depth-of-field and loss of focus inherent in macro-photography of uneven papyrus surfaces.

The training ran for 100 epochs using the NT-Xent loss function where $\tau = 0.5$ (temp.).

3) Mitigating Shortcut Learning: An unusual but key issue appeared when testing the SimCLR embeddings. During t-SNE visualization of the hidden space, two clear and widely split large clusters showed up. Still, these divisions did not align with character categories, authors, or historical styles.

After carefully checking the data details, we found an exact match between these groups and the library applied to create the crop.

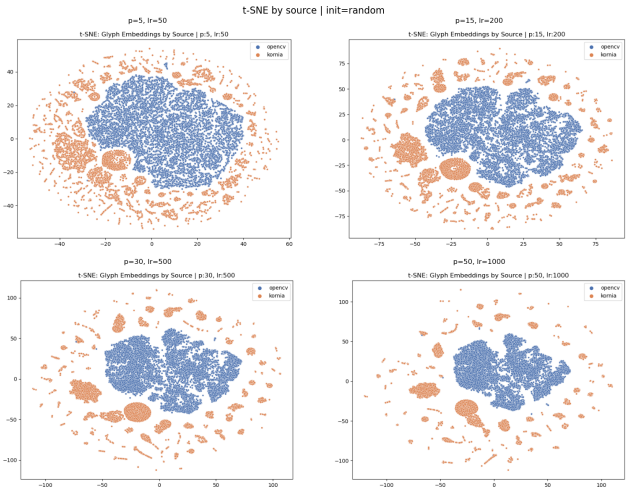


Fig. 3. Clusters generated with source bias

- Cluster A: Glyphs cropped using the `Kornia` GPU library.

- Cluster B: Glyphs cropped using the `OpenCV` CPU library.
- Analysis: Despite being configured for "Bi-linear Interpolation," subtle implementation differences in coordinate rounding along with anti-aliasing filters introduced tiny, high-frequency artifacts. The SimCLR model hooked into these hidden patterns as it was the best way to minimize contrastive loss.

Implication: This was fixed by making the whole process consistent by using "OpenCV" only throughout and remove the false sense of division.

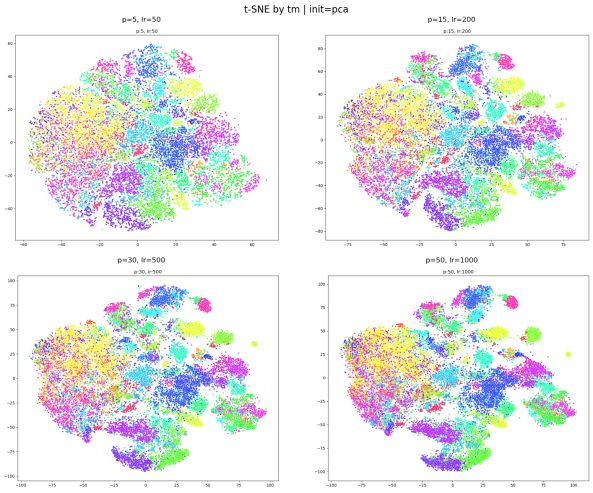


Fig. 4. Clusters generated after correction

B. Exploration 2: Supervised Contrastive Learning

To experiment with page-independent writer identification, we switched to a labelled approach for metric learning. This was an effectively better evaluation method to test out our approach.

1) Handling Noise without Binarization: Instead of using traditional methods like HOG, which usually need binarization to distinguish ink from paper, our deep learning model works directly on the images. Since the binarization might erase fine differences in stroke strength - features tied to individual handwriting. Moreover, the loss we are using should prevent the model from learning the background artifacts instead of the handwriting style.

2) Triplet Margin Loss and Sampling Strategy: We implemented a Triplet Margin Loss objective. The model is fed triplets (A, P, N):

- A (Anchor): A glyph from Writer X.
- P (Positive): A different glyph from Writer X (usually from a different page).
- N (Negative): A glyph from Writer Y.

The objective function minimizes the intra-class distance $d(A, P)$ while ensuring the inter-class distance $d(A, N)$ exceeds it by a margin α:

$$L = \max(0, \|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha) \quad (1)$$

A known challenge with Triplet Loss is that as the model improves, most random triplets satisfy the margin constraint ($L = 0$), leading to vanishing gradients and learning stops. To fix this, we implemented Batch-Hard Mining. Within each training batch B, for every anchor A, we select the *hardest* positive (the sample from the same writer that is farthest away) and the *hardest* negative (the sample from a different writer that is closest).

$$P_{hard} = \underset{P \in Batch}{\operatorname{argmax}}\, d(A, P) \quad (2)$$

$$N_{hard} = \underset{N \in Batch}{\operatorname{argmin}}\, d(A, N) \quad (3)$$

This ensures that the model focuses on the most unclear cases- such as distinguishing between two scribes with similar "Round" styles- thereby refining the decision boundaries more effectively than random sampling.

3) Network Architecture and Optimization: To further improve performance, we switched the backbone from ResNet18 to ResNet50, thereby significantly increasing the model capacity. We also added regularization techniques:

1) Projection Head with Dropout: The embeddings are passed through a MLP projection head with a Dropout layer ($p = 0.5$) before computing the loss.
2) Cosine Annealing Scheduler: The learning rate is slowly decreased along a wave-like pattern, which made it possible the system starts with major adjustments, then gradually stabilizes through incremental changes to a strong local minima.

C. Simple-VLAC

Because a single glyph gives little data, we would need to combine glyph features using pooling to create a page-wide representation. We utilized Simple-VLAC (Vector of Locally Aggregated Centers).

For each character class c (e.g., Alpha), we compute a global mean vector $\boldsymbol{\mu}_c$ from the training set. For a query page, we compute the sum of residuals across every occurrence of this character:

$$\mathbf{v}_c = \sum_{i=1}^{N_c} (\mathbf{x}_i - \boldsymbol{\mu}_c) \quad (4)$$

where $\mathbf{x}_i$ is the embedding of the i-th instance of character c on the page. These vectors are concatenated and power-normalized to form the final writer descriptor.

VI. Experimental Setup

The project was implemented using `python` and `PyTorch`.

A. Training Hyperparameters

For the final model, we tuned the hyperparameters to balance convergence speed and good generalization in the domain.

- Optimizer: We applied AdamW, with a weight decay set at $1e-4$. This helped control the projection head. It also prevented exploding weights during the training phase.

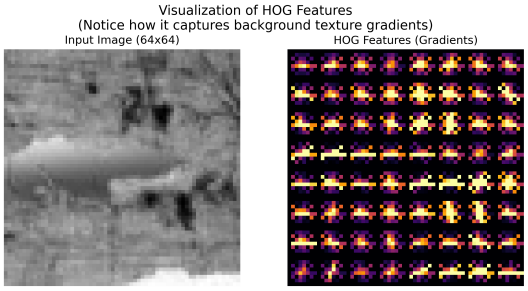


Fig. 5. Visualization of HOG features

- Learning Rate: To ensure the model could escape sharp local minima, we used a *Cosine Annealing* scheduler. We started with a rate of $1e-4$ and decayed it down to $1e-6$ over 300 epochs.
- Batch Size: We chose a batch size of 128 triplets. Not only does this utilize GPU capacity, but it also supports the *"Batch-Hard"* method more reliably. This increased the probability of finding truly difficult negative samples in each iteration.
- Margin (α): Based on tests, we chose a margin of 0.2 for Triplet Loss, as this value balanced cluster density and faster convergence.
- Input Resolution: Inputs were scaled to 64×64 pixels. We found this to be the computational "sweet spot", which was sufficiently detailed to preserve ink structure without incurring unnecessary computational overhead.

VII. Evaluation and Results

To ensure our results were reliable, we evaluated the pipeline on a strict "writer-disjoint" test split (20% of the dataset). In this case, the model never saw these specific writers during training, forcing it to learn generalized handwriting styles rather than simply memorizing the scribes.

A. Information Leakage in Handcrafted Features

To ensure reliability of our results, we performed a sanity check by benchmarking our approach against a classical computer vision baseline: Histogram of Oriented Gradients (HOG) aggregated with VLAD.

Unexpectedly, this simple baseline achieved an odd Top-1 accuracy of 99.99% on the test set. Given the degraded state of the documents, such a perfect score is not theoretically feasible and was a clear indicator of information leakage. At first we thought that the HOG descriptors were overfitting to the *background texture* of the papyrus fibers. Since fragments from the same manuscript share physical fiber patterns, the model probably recognized material traits instead of handwriting.

To validate this, we visualized the active HoG features from the image. As seen in Fig. 5, the gradients correlate with the vertical fiber patterns in the papyrus. This confirms the hypothesis.

Conclusion: The results showed this consistent signal wasn't due to visible image details, but instead reflected what we call "Digital Page Fingerprints." Instead of spotting obvious traits, the standard method picked up subtle repeating features - such as precise crop edges from YOLO, slight lighting shifts, or distinct noises in the image.

In contrast, our deep learning model (ResNet50) started with a much lower score ($\approx 24\%$) when tested without aggregation. This was actually a positive sign, as it confirmed that our Triplet Loss objective was not exploiting these easy "shortcuts." By forcing the model to pull together glyphs from *different* pages of a writer's corpus, it effectively penalized reliance on page-specific artifacts. Our model was confirmed as the only "true" learner.

B. Quantitative Results

After confirming the robustness of our feature encoder against artifact overfitting, we tested how the performance of our approach varies through optimization and aggregation strategies.

1) Efficacy of Page-Lebel Aggregation: Our initial evaluation with ResNet50 on single glyphs achieved 24.06% mAP (Table I). This confirms that individual glyphs, taken in isolation, are often too noisy or ambiguous to reliably identify a writer. However, applying Simple-VLAC resulted in better performance.

TABLE I
Impact of Model Architecture and Aggregation Strategy

Model	Evaluation Scope	mAP (%)
Unsupervised Baseline		
SimCLR + Linear Probe	Global (All Glyphs)	85.40%
Supervised ResNet18		
ResNet18 (Baseline, 100e)	Single Glyph	36.69%
ResNet18 (Optimized)	Global (All Glyphs)	69.80%
ResNet18 (Optimized)	Char-Specific (Alpha)	75.00%
ResNet18 (Optimized)	Char-Specific (Epsilon)	87.03%
Supervised ResNet50		
ResNet50 (Optimized)	**Page-Level (Simple-VLAC)**	**93.73%**

The refined pipeline achieved a state-of-the-art 93.73% mAP. This outcome matters greatly because it was achieved on a writer-disjoint split, proving the model generalizes to unseen hands. The jump from 24% to 93% demonstrates how the power of the law of large numbers work, i.e. while one glyph is noisy, the statistical distribution of a writer's style across a whole page is a unique fingerprint.

2) SimCLR Result: The SimCLR + Linear Probe achieved a high mAP of 85.40%. This showed that even without explicit supervision, the network learned robust features that correlate strongly with the writer's identity. However, this approach still required training a classifier on known writers, limiting its zero-shot utility.

C. Qualitative Evaluation: Embedding Space

To visually verify whether the model captured traits unique to each writer, we mapped the high-dimensional embedding

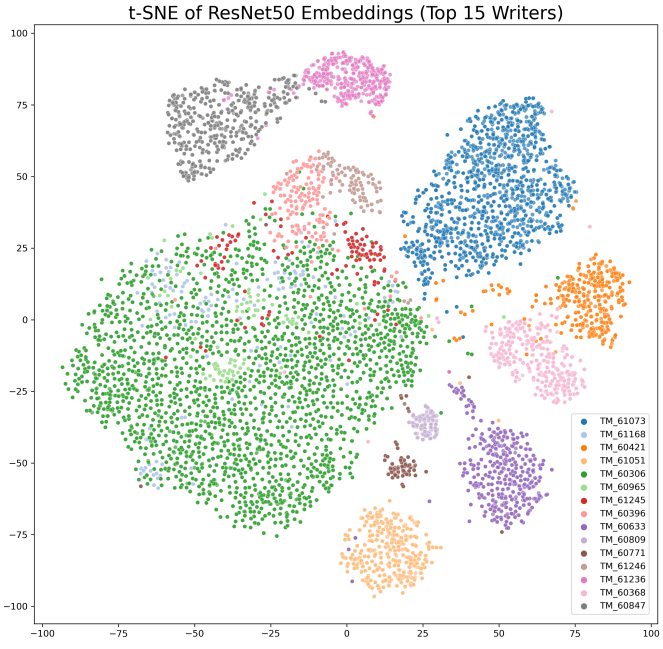


Fig. 6. t-SNE visualization of the learned embedding space.

space using t-SNE. As shown in Fig. 6, the plot reveals distinct, tight clusters that corresponds to individual writers *(TM IDs)*. Each color represents a unique writer (TM ID). The separation of clusters indicates that the model has learned a writer-specific representation, grouping glyphs from the same scribe even if they are in different character classes.

D. Error Analysis and Failure Modes

While the system achieved 93.73% mAP, a closer look at the failure cases provides insight into the limits of the current approach.

1. The ”Standard Style” Problem: The majority of confusion errors occurred between scribes writing in a highly formal ”Biblical Majuscule” or strict ”Koine” style. These styles are defined by how stiff they are (stroke pressure). If two writers follow them closely, exactly matching a strict standard, yet differing slightly in their personal movement patterns, they're nearly identical within the feature space.

2. Intra-Writer Variability: We observed that some scribes (e.g., TM_60220) show uneven patterns internally, resulting in glyphs that shift style significantly between the beginning and end of a document (possibly due to fatigue). In these cases, the global ”Mean Vector” used by Simple-VLAC fails to capture the variable nature of the writer's hand.

3. Rare Allographs: Errors were also more frequent when the query page contained rare allographs (stylistic variants) that wasn't repressented well in the training codebook. For example, if a test writer uses a cursive “loop“ Alpha that appears nowhere in the training set, the residuals calculated against the standard Alpha mean become uninformative noise vectors.

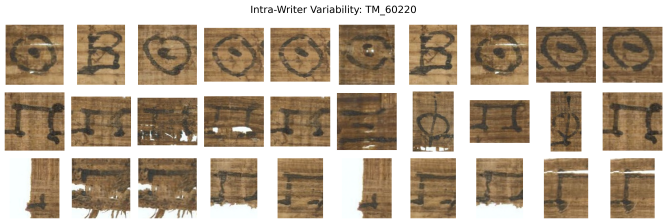


Fig. 7. Intra-Writer Variability: Scribe TM_60220

E. Cross-Dataset Generalization

To test robustness, we ran the model on an entirely external GRK Papyri 50 dataset in a zero-shot setting (without fine-tuning). This dataset contains 50 images from writers and manuscripts completely unrelated to the *HomerComp* training set.

Using Leave-One-Image-Out (LOIO) cross-validation, the model achieved a Top-1 Accuracy of 30.0%. While the score is lower than the intra-dataset score because of significant domain shift (differences in resolution and preservation), it far exceeds random predictions ($\approx 10\%$), indicating that the model picks up core patterns in handwriting that can be transferred across datasets.

TABLE II
ZERO-SHOT TRANSFER PERFORMANCE ON GRK PAPYRI 50

Metric	Top-1 Accuracy
Random Chance Baseline	$\approx$ 10.0%
Ours (Zero-Shot Transfer)	**30.0%**

As shown in Table II, the system reached a Top-1 Accuracy of 30.0% (15/50 pages).

The performance gap shows how hard it is to handle domain shift in digital papyrology. Due to variations in scanning quality, condition of documents, or writing styles across the HomerComp and GRK sets, visual traits differ noticeably. Still, getting the right scribe about a third of the time - without adjusting the model at all - suggests it is learning core stroke patterns instead of memorizing data quirks.

VIII. QUANTITATIVE PALEOGRAPHY

We shifted from simple binary classification to test whether our deep embeddings could function as forensic tools. By projecting writers into an embedding space, we found out that we could quantify abstract concepts like ”uniqueness” or ”consistency,” effectively bridging the gap between computational metrics and humanities research.

A. Assessing Stylistic Distinctiveness

A key question in paleography is determining how much a scribe deviates from the standard writing norms of their time. To measure this factor of ”idiosyncrasy,” we computed the cosine distance between a specific writer's average vector (focusing on the highly informative *Epsilon*) and the global average *Epsilon* derived from the entire dataset.

This measure makes it possible to objectively rank scribes by the “divergence“ of their hand. As shown in Fig. 8, the analysis identifies specific scribes (e.g., `TM_60734`) as statistical outliers. These writers have highly unique stylistic signatures, whereas writers at the bottom of the ranking conform closely to the standard “Koine“ style of the period.

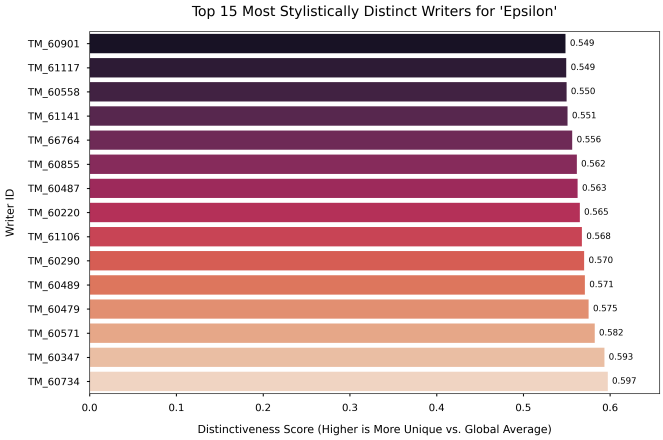


Fig. 8. Ranking writers by stylistic distinctiveness ’Epsilon’.

B. Writing styles: Consistency vs. Distinctness

We also examined scribal patterns by plotting two dimensions of style:

1) Intra-Letter Cohesion (y-axis): How similar a writer’s glyphs are to their average value. A high score indicates a highly disciplined hand with low motor variance.
2) Inter-Letter Distance (x-axis): How distinct a writer’s characters are from one another (e.g., the distance between their ’Alpha’ and ’Epsilon’). A high score indicates high physical structure and clarity.

Fig. 10 shows distinct clusters of scribal behavior:

- Clear & Neat (Top-Right): Writers that show clarity along with control consistent yet keeping unique character shapes.
- Outlier but Distinct (Bottom-Right): Writers with high varying style (low consistency) but whose character topologies remain distinct enough to be legible.
- Ambiguous (Top-Left): Writers who are consistent but whose letter forms resemble each other, leading to potential ambiguity.



Fig. 9. Handwriting comparison between different writers

IX. Future Work

Various paths could improve the system.

The immediate next best step would be to binarize the images before training. Initially this was persumed to be redundant, since Triplet Margin Loss prevents learning the background details, which is the objective of binarizing the image. This would require training a U-Net for further processing

X. Conclusion

This work successfully established a robust, end-to-end pipeline for extracting and identifying ancient Greek handwriting. However, continuing with the project development revealed that in the Digital Humanities, cleaning data is often more important than model architecture. Standard benchmarks can be deceptively high if digital artifacts like sensor noise are not properly accounted for.

We showed that while traditional methods like HOG easily overfit to these ”page fingerprints”, a deep metric approach (specifically Triplet Loss) pushes the model to capture genuine stylistic features. By combining an optimized ResNet50 with our character-specific Simple-VLAC aggregation, we achieved a final mAP of 93.73%.

Ultimately, this project suggests that deep embeddings are not just black boxes for classification. They function as quantitative tools for paleographers, offering objective metrics for concepts like scribal distinctiveness that were previously left to subjective interpretation by experts.

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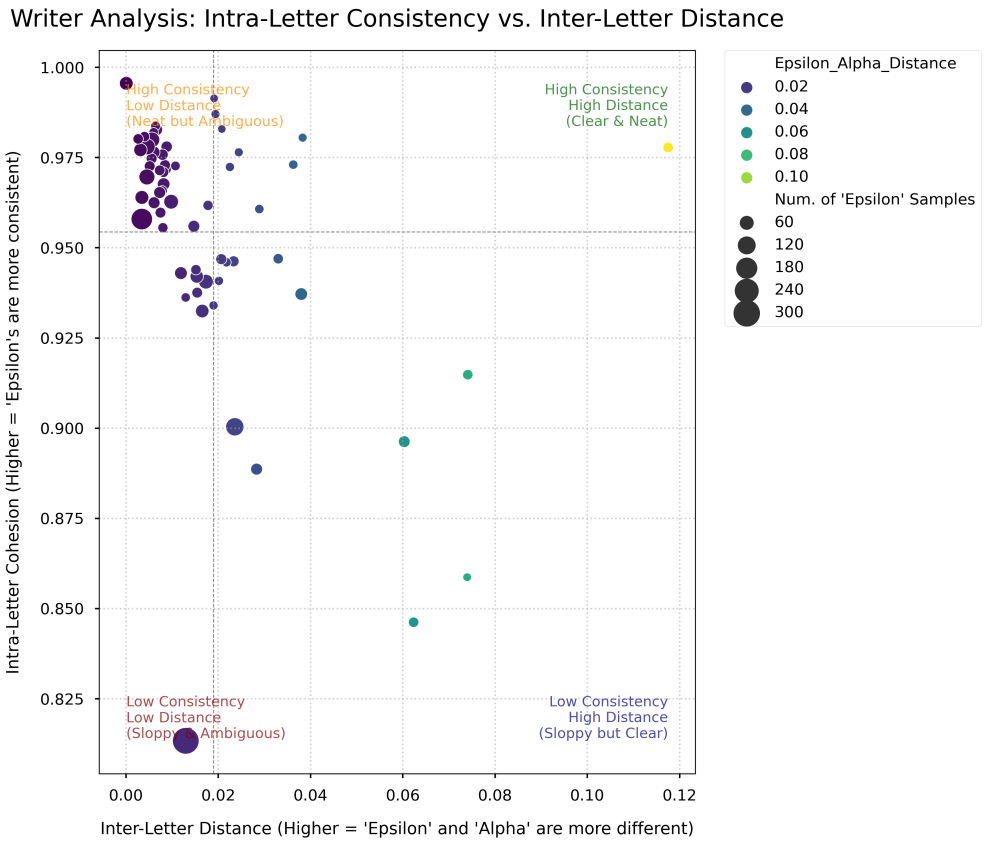


Fig. 10. Mapping scribal habits based on consistency (y-axis) and letter distinctness (x-axis).

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